

SECTION 9 – PREINSTALLATION CHECKLIST

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9-1 INTRODUCTION

“Preinstallation” refers to work necessary to plan and prepare a site for delivery and installation of equipment. Delay, confusion, and waste of manpower can be avoided by completing preinstallation work. It is recommended to have GE Service Representative make on-site inspections during construction.

The purpose of this section is to outline key areas of concern in the preparation of a customer's site for magnetic resonance. It is intended as a guide for GE's Installation Specialists and/or Installation Leaders when making on-site inspections during the construction phase of an MR project. Note that these inspections by GE are intended to aid the overall site preparation process. They do not relieve the customer and project architect from responsibility for the design, engineering, and coordination efforts necessary to ensure a successful MR project.

During the course of the site preparation process, GE's Installation Specialists and/or Installation Leaders may observe that the requested arrival date (RAD) for the MR equipment is not realistic. If this is the case, appropriate actions must be taken by the GE Field Sales/Service Team to adjust the RAD accordingly.

All work must be in compliance with national and local safety codes.

9-2 GENERAL PREINSTALLATION REMINDERS

- ☐ 1. Has the customer's architect fully reviewed final site construction drawings using the guidelines provided by MR Siting?
- ☐ 2. Has the vibration study been completed per **Section 4-15, VIBRATION**? It is the customer's responsibility to contract a vibration consultant or qualified engineer to implement design modifications to meet the specified limits.
- ☐ 3. If there were vibration issues; has local GE Field Service verified, with the vibration consultant present, the elimination/reduction of all identified sources?
- ☐ 4. Have vehicle parking arrangements been made for installation personnel?
- ☐ 5. Is temporary storage space available for use during installation?
- ☐ 6. What is facility smoking policy?
- ☐ 7. Have facility arrangements been made for refuse disposal during installation?

9-3 SAFETY

WARNING!

THE FOLLOWING SAFETY ITEMS MUST BE ADHERED TO PRIOR TO MOVING THE MAGNET FROM THE TRUCK INTO THE ROOM. THE MAGNET WILL BE EXHAUSTING HELIUM GAS DURING THIS TIME. GASEOUS HELIUM IS AN INVISIBLE, ODORLESS GAS THAT CAN CAUSE ASPHYXIATION WHEN DEPLETING THE AMBIENT OXYGEN SUPPLY. HELIUM GAS IS LIGHTER THAN AIR AND WILL RISE TO THE CEILING.

- ☐ 1. Have all of the required room ventilation items been installed and tested to make sure sufficient ventilation is available? Is exhaust fan and fan control installed and functional? The customer is to supply a copy of the Exhaust Fan Test Report to GE Medical Systems for the customers record file. Refer to **Section 4-8 ROOM VENTILATION** for specific requirements.
- ☐ 2. Are a first aid kit and non-ferrous fire extinguishers available at site?
- ☐ 3. Are removable chain or strap available for securing cryogen gas cylinders in an upright position to prevent the cylinders from falling which may cause injury or damage?
- ☐ 4. Are functioning telephones as defined in **Section 2-8-8 System Monitoring & Support Connectivity Requirements** available at site for duration of installation? Phone lines are needed to dial out in case of an emergency.
- ☐ 5. Has the cryogenic vent been installed complete from the Magnet Room RF shield penetration to the final exit and been inspected to final exit outside of the building? The customer is to supply a copy of the Vent Inspection Report to GE Medical Systems for the customers record file. Refer to **Section 4-9, CRYOGENIC VENTING**.
- ☐ 6. Have plans been made to connect the cryogenic vent to the magnet after magnet installation in the room? Refer to **Section 4-9, CRYOGENIC VENTING**.
- ☐ 7. Is there adequate ventilation for helium dewar storage area. Refer to **Section 4-8, ROOM VENTILATION**.

9-4 PREPARATIONS REQUIRED IN ADVANCE OF MAGNET DELIVERY AND MOVING INTO MAGNET ROOM

The following items must be completed prior to magnet delivery. A site inspection by GE Service Representative must be completed prior to magnet delivery to ensure site readiness.

- ☐ 1. Has Main Disconnect Panel been installed, electrician wiring complete, MDP operational with power available? Refer to **Table 5-1 REQUIRED CUSTOMER POWER**.
- ☐ 2. Is 24 hour/day, 7 days/week power and air cooling available for Main Disconnect Panel (MDP) for powering the Outdoor or Indoor Twin System Cooling Cabinet (TSCC), Shield/Cryo Cooler Compressor cabinet, and Magnet Monitor equipment?
- ☐ 3. If MR system configuration is utilizing water cooled TSCC then is 24 hour/day, 7 days/week water cooling available?

9-4 PREPARATIONS REQUIRED IN ADVANCE OF MAGNET DELIVERY AND MOVING INTO MAGNET ROOM (Continued)

- ☐ 4. Are all walls and ceiling in magnet room essentially complete except for removable section?
- ☐ 5. Has a clear route to the magnet room been defined for magnet installation (refer to **Section 2-4, MINIMUM DOOR SIZES**)?
- ☐ 6. Is a secure space available to store equipment on site?
- ☐ 7. Have arrangements been made for the use of special rigging equipment for moving the magnet into the magnet room?
- ☐ 8. Has the raised floor been installed in the equipment/computer room in preparation for the superconducting and resistive power supplies?
- ☐ 9. Has work in the magnet room been completed or suspended and the magnet room closed off to provide a dust-free, closed environment?
- ☐ 10. Has all equipment been removed from the magnet room to allow space for the magnet with rigging equipment and cryogen dewars?
- ☐ 11. Are equipment anchors installed in floor as required for magnet room equipment? Refer to **Section 7-7 ANCHOR SPECIFICATION FOR MR EQUIPMENT INSIDE RF SHIELDED ROOM**.
- ☐ 12. For all magnet room anchors, has pull test been performed and installation torque value defined? Test results recorded by RF Shield Room Vendor and information forwarded to GE Medical Systems for the customers record file? Refer to **Section 7-7-1 Clamping Force And Pull Test**.
- ☐ 13. For all magnet room anchors, has ground impedance test been performed? Test results recorded by RF Shield Room Vendor and documentation given to GE Medical Systems for the customers record file? Refer to **Section 7-7-3 Electrical Isolation**.

9-5 PREPARATIONS REQUIRED IN ADVANCE OF CRYOGEN TRANSFER/MAGNET FILL

- ☐ 1. Have arrangements been made for cryogen delivery and storage?
- ☐ 2. Does a clear path exist to magnet room for delivery of cryogens?
- ☐ 3. Is the magnet room adequately vented to allow exhaust of cryogen boil-off?
- ☐ 4. Is the magnet room clear to provide room for cryogen dewars during filling?

9-6 PREPARATIONS REQUIRED IN ADVANCE OF SYSTEM DELIVERY/INSTALLATION

The following items must be completed prior to system delivery. A site inspection by GE Service Representative must be completed prior to system delivery to ensure site readiness.

- ☐ 1. Has the equipment/computer room raised floor been installed if not previously completed?
- ☐ 2. Has area under the raised floor been vacuum cleaned and free of all debris?
- ☐ 3. Have any necessary conduits or raceway for power cables been installed?
- ☐ 4. Has delivery route been defined for equipment (refer to **Section 2-4, MINIMUM DOOR SIZES**) including plans for protecting flooring for heavy equipment & carts (ie. RF/Gradient Body Coil & shipping cart)?
- ☐ 5. Is power available for connection to the Main Disconnect Panel?
- ☐ 6. Does incoming power have all the specified safety precautions and remote disconnects?
- ☐ 7. Is the operator's area complete to provide a dust-free environment for installation of the Operator Workspace equipment?
- ☐ 8. Are the areas for optional equipment complete to provide a dust-free environment for installation of equipment?
- ☐ 9. Is air conditioning running in equipment room?
- ☐ 10. Are functioning telephones as defined in **Section 2-8-8 System Monitoring & Support Connectivity Requirements** available at site for duration of installation?

9-7 PREPARATIONS REQUIRED IN ADVANCE OF MAGNET RAMP-UP

- ☐ 1. Is the cryogenic vent connected to the magnet?
- ☐ 2. Has the magnet room been completely closed (removable section closed up and sealed)?
- ☐ 3. Has the magnet room been tested to ensure that the RF shielding meets the electrical isolation and the 100dB requirement for electric/plane waves and the 90dB requirement for magnetic waves refer to **Section 7-1 RF SHIELDED ROOM INTRODUCTION**? The customer is to supply a copy of RF shielded room vendor test reports.
- ☐ 4. Have all ferrous metal objects been removed from the magnet room?
- ☐ 5. Have adequate signs (Safety and Exclusion Zones) been posted to warn personnel about dangers of magnetic field?
- ☐ 6. Have facility personnel been informed of magnet safety precautions and procedures?
- ☐ 7. Has power been connected from the Main Disconnect Panel to Power Distribution Unit?
- ☐ 8. Has the penetration panel been installed?
- ☐ 9. Is all contractor construction work completed?
- ☐ 10. Has all contractor equipment that could affect shimming been removed from within the three gauss zone?
- ☐ 11. Have precautions been taken to prevent movement of large metal objects within the three gauss zone?
- ☐ 12. Have local fire department(s) and police department(s) been informed of unique characteristics (e.g. strong magnetic field, cryogenics, etc.) of magnet and correct precautions to take in event of emergencies?

9-8 TYPICAL MR INSTALLATION PROJECT SCHEDULE

The MR Project schedule shown below depicts a typical Signa TwinSpeed installation. The purpose of this project schedule is to provide a guide for your MR project in regards to tasks which need to be completed, duration of each task and who is primarily responsible for completion of the task. This project schedule starts at MDP and TSCC/or magnet delivery which occurs when your site construction is completed, except for magnet room opening. Your GE Installation Specialist can provide more detailed information if needed.

TYPICAL MR INSTALLATION PROJECT SCHEDULE

Typical MR Installation Schedule	Duration	Responsibility	Week 1					Week 2					Week 3					Week 4					Week 5				
			M	T	W	T	F	M	T	W	T	F	M	T	W	T	F	M	T	W	T	F	M	T	W	T	F
GE MDP & TSCC or TGWC equipment delivered 1 week prior to magnet delivery	0.25 day	Installation Specialist (IS) & Contractor	MDP & TSCC/TGWC must be installed & operational prior to magnet delivery																								
MDP Installation	0.5 day	IS & Contractor																									
Outdoor TSCC/TGWC Installation	1 day	IS & Contractor																									
Indoor TSCC/TGWC Installation	0.5 day	IS & Contractor																									
Magnet delivered	0.5 day	GE																									
Magnet mechanical / electrical	0.5 day	GE																									
Magnet room closure	3 days	Contractor																									
Shielding is completed and tested		Contractor																									
Complete magnet room opening		Contractor																									
Flooring completed		Contractor																									
MR system delivered	0.5 day	GE																									
Mechanical Installation of system	3 days	GE																									
Calibration of system	12 days	GE																									
MR system available to customer																											